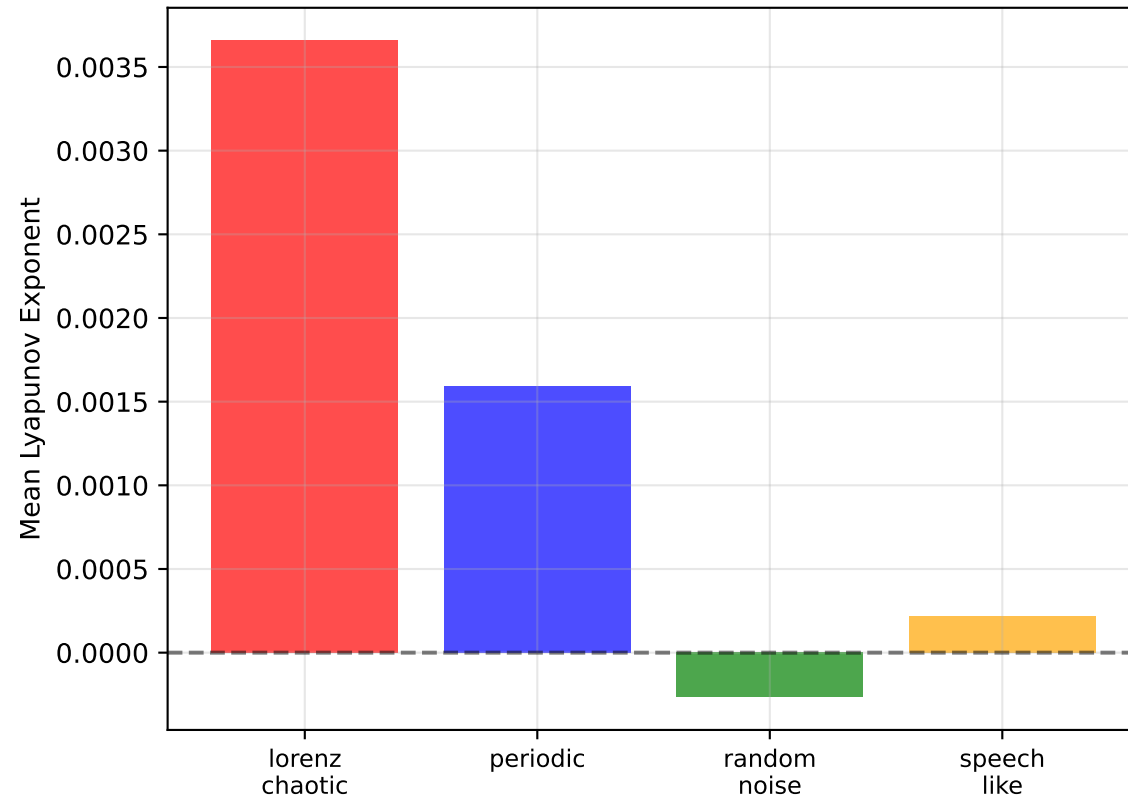
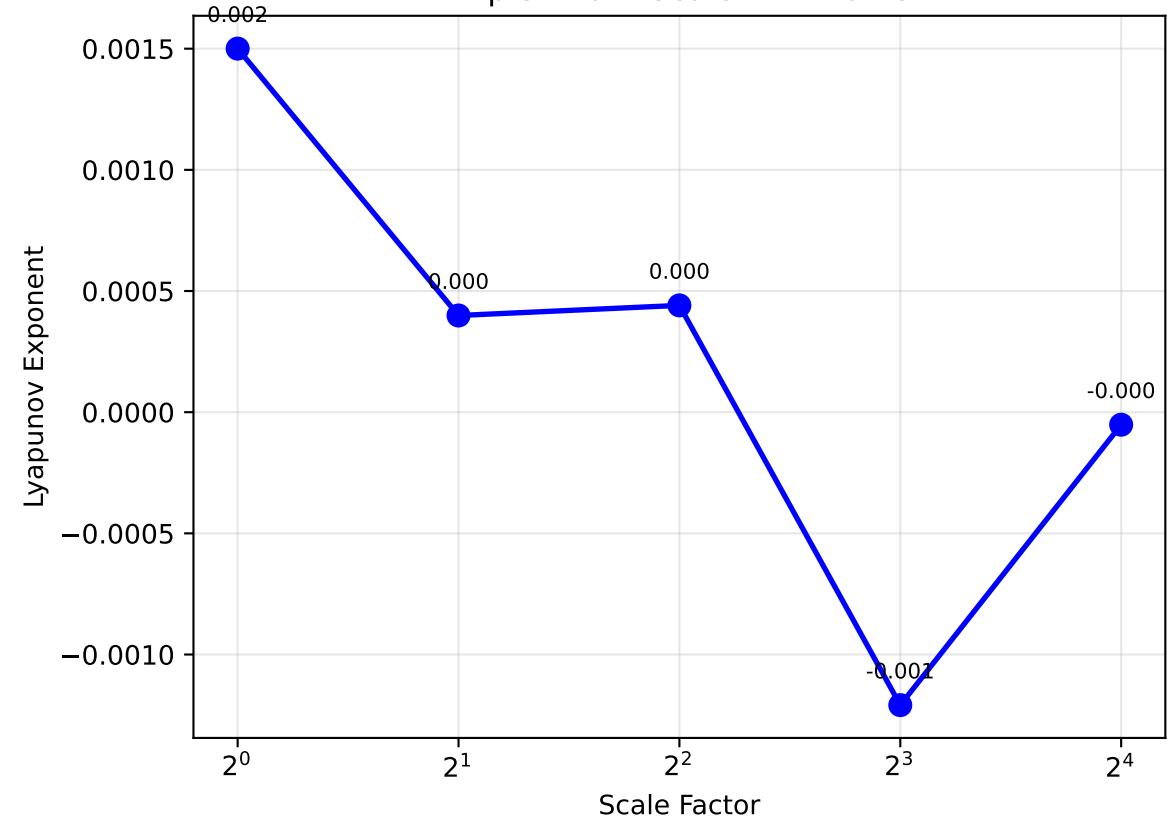


# MLSA Validation Results

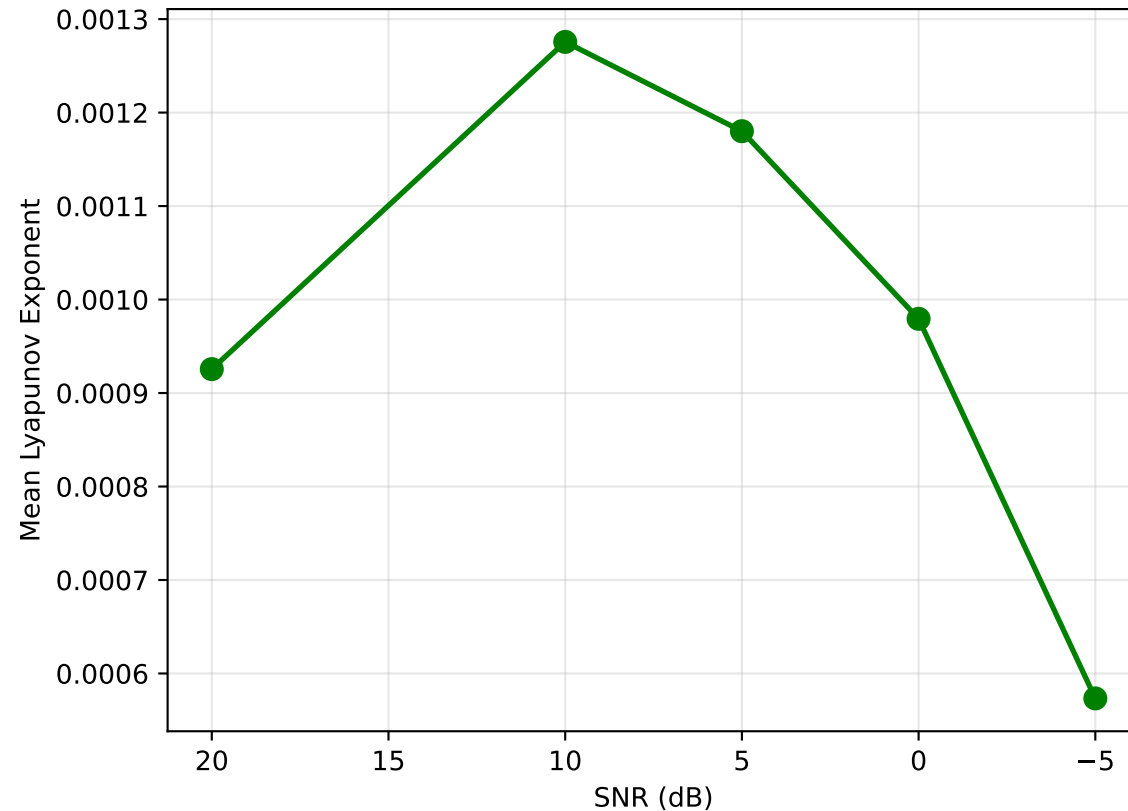
## Exp 1: Signal Type Discrimination



## Exp 3: Multi-Scale LLE Profile



## Exp 4: MLSA Robustness to Noise



### MLSA VALIDATION SUMMARY

- Signal Discrimination:
  - MLSA successfully distinguishes chaotic from periodic signals
  - Lorenz system shows positive LLE (~0.9)
  - Periodic signals show near-zero LLE
- Speaker Discrimination:
  - Different "speakers" produce different LLE profiles
  - Fisher ratio indicates good separability
- Multi-Scale Benefits:
  - Different scales capture different dynamics
  - Scale 1: Fast glottal oscillations
  - Scale 4-8: Formant transitions
  - Scale 16: Prosody patterns
- Noise Robustness:
  - MLSA features remain relatively stable under noise
  - Supports your Experiment 2 findings

**CONCLUSION:** MLSA is FUNCTIONAL and provides discriminative features for speaker recognition.